

5. (a) The table shows a comparison for two athletes, Tomos and Jeremy, who ran in races of different distances.

Athlete	Distance of race (m)	Oxygen needed in the race (dm ³)	Oxygen entering blood in the race (dm ³)
Tomos	100	10	0.5
Jeremy	10 000	150	134.0

- (i) The difference between the oxygen needed and the oxygen actually entering the blood during the race is the oxygen debt. Calculate the oxygen debt for [1]

Tomos dm³

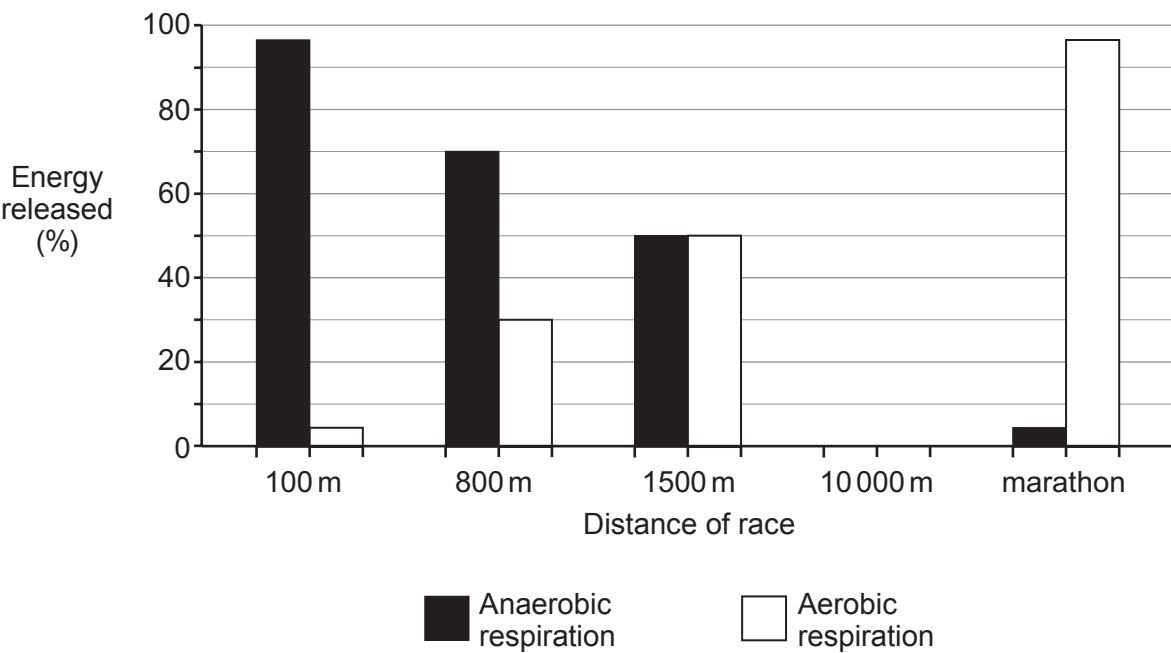
Jeremy dm³

- (ii) When the race was over, both athletes continued to breathe more rapidly and more deeply than normal for some time. Give the reason for this. [1]

.....

.....

- (b) The bar chart shows the percentage of energy released by aerobic and anaerobic respiration for athletes running different distances.



Examiner
only

- (i) What can be concluded from the data shown in the table and the bar chart for the athlete running 100 m? [1]
-
-
- (ii) **On the bar chart opposite, draw bars** that you would expect for an athlete running in a 10 000 m race. [2]
- (c) Write the word equations to represent:
- (i) Aerobic respiration in human muscle tissue; [1]
-
- (ii) Anaerobic respiration in human muscle tissue. [1]
-
- (d) Why is aerobic respiration more efficient than anaerobic respiration? [1]
-
-

3400UA01
11

8